$$= \frac{\frac{3}{4} - \frac{1}{4}}{\sqrt{5 \times \frac{1}{4} \times \frac{3}{4}}}$$

$$= \frac{\frac{2}{4}}{\sqrt{\frac{15}{16}}}$$

$$= \frac{2}{\sqrt{15}}$$

Finding: The distribution is positively skewed.

Kurtosis

$$\text{Kurtosis} = \frac{1 - 6pq}{npq}$$

$$= \frac{1 - 6 \times \frac{1}{4} \times \frac{3}{4}}{\frac{15}{16}}$$

$$= \frac{\frac{-2}{16}}{\frac{15}{16}}$$

$$= \frac{-2}{15}$$

$$= -0.1333$$

Finding: The distribution is Platykurtic.

Example 10.6

In a Binomial distribution with 7 trials, $P(X=3)=P(X=4)$ Check whether it is a symmetrical distribution?

Solution:

A Binomial distribution is said to be symmetrical if $p= q= \frac{1}{2}$

Given: $P(X=3) = P(X=4)$

$X \sim B(n, p)$

$P(X = x) = {}^{n}C_{x}\, p^{x}\, q^{n-x}$, $x= 0,1,2,\ldots n$

${}^{n}C_{3}\, p^{3} q^{n-3} = {}^{n}C_{4}\, p^{4}\, q^{n-4}$

${}^{7}C_{3}\, p^{3} q^{4} = {}^{7}C_{4}\, p^{4} q^{3}$ note that ${}^{7}C_{3} = {}^{7}C_{4}$

On simplifying, we have $q = p$